GP Meeting 629

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1 Introduction

The current experiments focus on understanding when and why the single-band Gaussian Process (GP) model becomes inaccurate, overconfident, or underconfident. Beyond comparing performance by astrophysical class, I also investigate object-level light-curve features and their relationship with GP evaluation metrics. This is useful because class-level differences may be confounded with data quality, sampling density, cadence, signal-to-noise ratio, or intrinsic curve morphology.

1.1 Number of Objects Control

One possible experiment is to fix the number of objects per class to be the same. This would make class-level comparisons more balanced, because classes with more objects naturally have more stable aggregated estimates.

However, this is not necessarily required for the main evaluation. The single-band GP is fitted independently to each object, and there are no shared parameters across objects. Therefore, the number of objects in a class does not affect the GP prediction for an individual object. It only affects the statistical reliability of the class-level aggregated metrics.

For this reason, the main analysis can use all available objects per class, while reporting the number of objects together with each metric. Equal-object subsampling or bootstrapping can be added later as a robustness check.

1.2 Effect of Light-Curve Features on Calibration

The first analysis is performed at the class level. For each class, I compute the average light curve per band and visually inspect the characteristic morphology, as shown in Figure 1. This shows clear class-level differences. Supernova-like classes usually have a sharp peak and relatively flat regions away from the peak, while classes such as AGN, RRL, Mira, or EB can show more irregular, oscillatory, or chaotic-looking behavior.

However, class-level analysis has an important limitation. The number of objects per class is not fixed, so the aggregated results do not have the same variance and reliability. Classes with more objects produce more stable estimates,

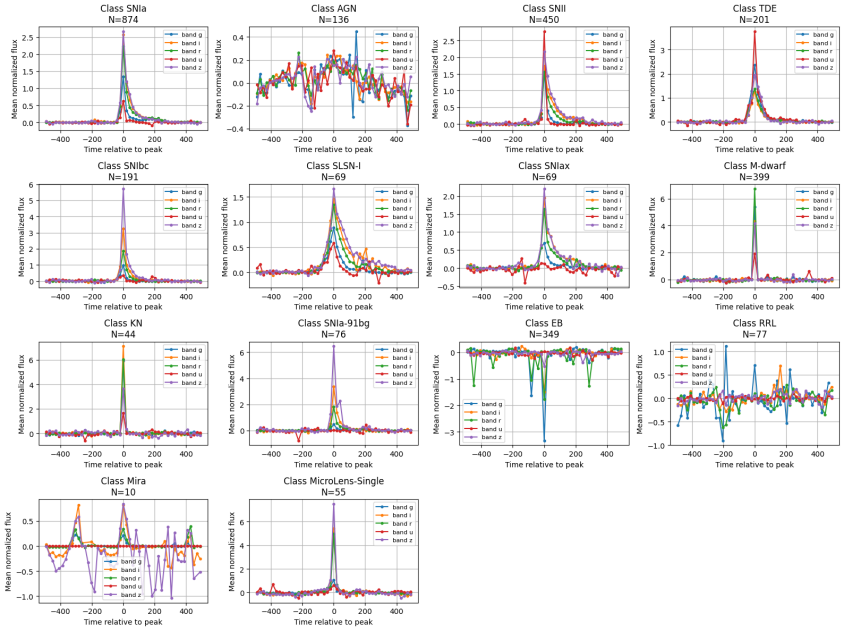


Figure 1: evolution of the mean normalized flux per band over objects from the same class.

while classes with fewer objects may have much larger uncertainty. In addition, class effects may be confounded with data quality. For example, a class may appear badly calibrated not because its intrinsic light curves are harder, but because it has fewer observations, poorer cadence, lower signal-to-noise ratio, or more extrapolated test points.

The second possible analysis is to examine a single class only, especially SNIa, which has the largest number of objects. This would reduce class-level confounding and allow us to study how different feature regimes affect GP performance within one astrophysical class. This experiment is not yet complete.

The third and currently most useful analysis is feature-based. Instead of grouping objects only by class, I compute object-level or object-band-level features from several categories: morphology, fluctuation or roughness, signal-to-noise ratio, and sampling geometry. I then evaluate GP performance at the object-band level and analyze how the evaluation metrics vary with these features.

This approach is often more scientifically interpretable than only comparing classes, because it asks directly under which data or light-curve conditions the GP performs poorly.

2 Experiment on Single-Band GP

2.1 PIT histogram and reliability curve

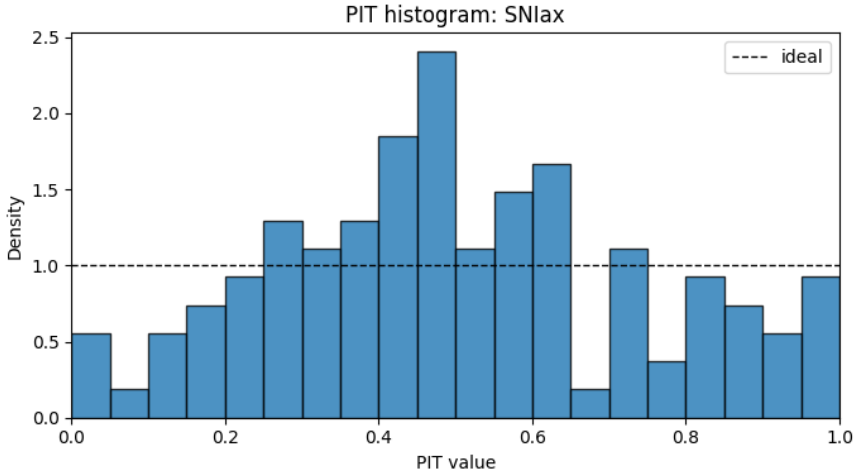
First, the PIT histograms and their corresponding PIT reliability curves are plotted for all classes to inspect the calibration behaviour of the single-band GP model. For a well-calibrated probabilistic model, the PIT values should approximately follow a uniform distribution on $[0, 1]$, and the empirical PIT reliability curve should lie close to the diagonal. A concentration of PIT values around 0.5 indicates that the predictive distributions are too wide, i.e. the model is underconfident. In contrast, a U-shaped PIT histogram, with too many PIT values close to 0 and 1, indicates that the predictive distributions are too narrow, i.e. the model is overconfident.

For the SNIax class, the PIT histogram shows a clear concentration around 0.5 and a depletion near both tails. The PIT reliability curve lies below the diagonal for small probabilities and above the diagonal for large probabilities, forming an S-shaped deviation. This indicates that the predictive distributions are underconfident: the held-out observations fall too often near the centre of the predictive CDF. This observation is consistent with $z_\mathrm{std} < 1$. The small asymmetry of the histogram and reliability curve also suggests that a weak bias component may be present.

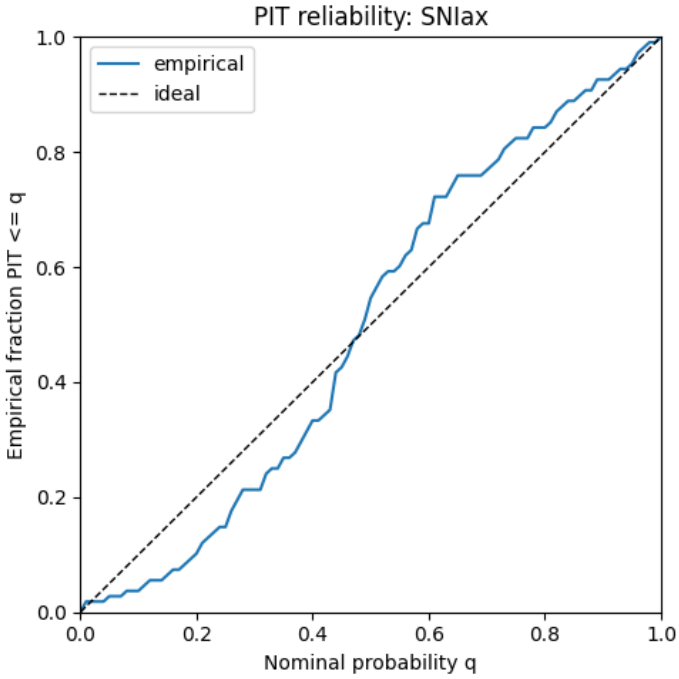
A similar pattern is observed for most supernova-like classes, including SLSN-I, SNII, SNIa, SNIa-91bg, SNIbc, and TDE. Their PIT histograms are generally concentrated around the middle of the unit interval, while the reliability curves show the same S-shaped structure: below the diagonal for low q and above the diagonal for high q. This is consistent with their average standardized residual spread, since these classes mostly have $z_\mathrm{std} < 1$. Hence, the GP predictive variances are often too large compared with the actual held-out residuals. The effect is especially clear for SLSN-I, SNIa-91bg, SNIax, SNIbc, and TDE, where the central concentration of PIT values is visually strong.

The non-supernova classes show more heterogeneous behaviour. KN is relatively noisy because it contains only a small number of objects, so its PIT histogram and reliability curve should not be over-interpreted. RRL shows the opposite tendency from the supernova-like classes: its PIT histogram has strong mass near the two boundaries and its reliability curve is not S-shaped in the same way. This suggests overconfidence, which is also consistent with its large value of $z_\mathrm{std} > 1$. However, this result is based on only five RRL objects, so the conclusion is less reliable. AGN and EB also do not follow the simple underconfident pattern. Their PIT distributions contain substantial tail mass, and their z_std values are larger than one, suggesting that for these classes the GP uncertainty can be too narrow or affected by outliers and model mismatch. M-dwarf is visually closer to the underconfident pattern because its PIT histogram has a strong central peak, but its class-level z_std has high variability, so the aggregated result may be affected by a few poorly fitted objects.

Overall, the PIT diagnostics suggest that the dominant calibration issue for the supernova-like classes is underconfidence, or overdispersion of the predictive



(a) PIT histograms for different object classes.



(b) PIT reliability curves for different object classes.

Figure 2: PIT diagnostics for all object classes. (a) PIT histograms. A uniform histogram indicates good calibration, while a central concentration indicates underconfidence and excess mass near 0 and 1 indicates overconfidence. (b) PIT reliability curves. The dashed diagonal represents ideal calibration. An S-shaped deviation, with the empirical curve below the diagonal for small q and above the diagonal for large q, indicates underconfident predictive distributions.

distributions. The model tends to assign too much uncertainty to held-out observations, causing the realized PIT values to cluster around 0.5. This agrees with the class-level standardized residual statistics, where most supernova-like classes have $z_\mathrm{std} < 1$. In contrast, some non-supernova classes, especially RRL, AGN, and EB, show signs of overconfidence or more complex miscalibration. Therefore, the calibration error is not uniform across classes: it depends on the object type and likely reflects differences in light-curve morphology, sample size, variability pattern, and the suitability of the single-band Matérn GP model for each class.

2.2 Evaluation Setup

The single-band GP is evaluated independently for each object and "r" band. For each evaluated unit, the feature table contains metadata, extracted light-curve features, and GP evaluation metrics.

The GP metrics are computed in **normalized flux space**. This is important because raw-space RMSE, CRPS, and NLPD are scale-dependent and can be affected by object brightness or flux amplitude. Normalized-space metrics are more suitable for comparing GP performance across different objects and classes.

The main metrics considered are:

- RMSE: point prediction error;
- CRPS: probabilistic prediction quality, combining accuracy and uncertainty;
- NLPD: negative log predictive density;
- `z_std`: standard deviation of standardized residuals;
- `z_error`: signed calibration error, defined as $z_{\mathrm{error}} = z_{\mathrm{std}} - 1$;
- empirical coverage at one, two, and three sigma;
- KS-PIT: deviation of the PIT distribution from uniformity.

The interpretation of `z_error` is:

$$z_{\mathrm{error}} > 0 \quad \Rightarrow \quad \text{overconfident GP},$$

Conversely,

$$z_{\mathrm{error}} < 0 \quad \Rightarrow \quad \text{underconfident GP}.$$

2.3 Essential Features

The extracted features are divided into four main groups.

2.3.1 Morphology Features

Morphology features are computed from the complete observed light curve. They describe the overall shape of the curve. Flux is normalized before computing these features.

Important morphology features include:

- `flat_fraction`: fraction of points close to zero relative to the object's amplitude;
- `width_50_pos`: approximate width of the positive peak above 50% of its amplitude;
- `flux_range_robust`: robust vertical flux range, defined using the 95th and 5th percentiles;
- `amp_abs`: maximum absolute normalized amplitude.

These features are designed to distinguish transient-like curves from irregular or oscillatory curves. For example, supernova-like curves often have high flat fraction and a well-defined peak, while variable sources may have lower flat fraction and more persistent variability.

2.3.2 Fluctuation and Roughness Features

Fluctuation features describe how rapidly and irregularly the light curve changes over time. These are also computed using normalized flux.

Important features include:

- `mean_abs_slope`: average absolute time-normalized slope;
- `total_variation`: sum of absolute flux differences along the time-ordered curve;
- `median_crossing_rate`: how often the curve crosses its median level;
- `slope_std`: variability of local slopes.

These features quantify the visual impression of chaotic or rapidly varying curves. In the experiments, they are especially relevant to CRPS.

2.3.3 Signal-to-Noise Features

Signal-to-noise features are computed from the original unnormalized flux and flux errors. This is necessary because signal-to-noise ratio measures the relation between the physical flux scale and the measurement uncertainty.

Important features include:

- `peak_snr`: peak signal-to-noise ratio;
- `noise_to_signal`: typical flux error divided by the robust flux scale;

- `median_flux_err`: median original flux uncertainty.

A high `noise_to_signal` means that the observational noise is large relative to the signal scale. This can make the curve appear irregular and can degrade probabilistic prediction performance.

2.3.4 Sampling and Train-Test Features

Sampling features are computed from the same training and test split used in the original single-band GP evaluation. No new split is introduced during feature extraction.

Important features include:

- `n_train`: number of training points;
- `median_train_cadence`: median time interval between training observations;
- `gap_90_train`: 90th percentile of training gaps;
- `nearest_train_distance_mean`: average distance from each test point to its nearest training point;
- `fraction_train_near_peak`: fraction of training points near the peak region.

These features describe what the GP actually observed and how difficult the evaluation problem is. For example, a large nearest-train distance means that test points are poorly supported by nearby training observations.

2.4 Correlation Among Features

To avoid manually checking every possible pair of variables, I use Spearman rank correlation as an initial screening tool. Spearman correlation is useful because it detects monotonic relationships and is less sensitive to nonlinear scaling than Pearson correlation.

The feature correlation matrix shows that many features are strongly correlated and should not be interpreted as independent causes.

Table 1: Spearman correlation matrix among roughness and noise-related features.

	mean_abs_slope	total_variation	flux_range_robust	median_crossing_rate	noise_to_signal	peak_snr
mean_abs_slope	1.000000	0.915012	0.808970	0.500441	0.795507	-0.538702
total_variation	0.915012	1.000000	0.779933	0.451011	0.738689	-0.395663
flux_range_robust	0.808970	0.779933	1.000000	0.367877	0.810778	-0.641376
median_crossing_rate	0.500441	0.451011	0.367877	1.000000	0.375787	-0.209241
noise_to_signal	0.795507	0.738689	0.810778	0.375787	1.000000	-0.817007
peak_snr	-0.538702	-0.395663	-0.641376	-0.209241	-0.817007	1.000000

In particular, the roughness-related features are highly redundant:

- `mean_abs_slope` and `total_variation` have a very strong correlation;
- `flux_range_robust` is also strongly correlated with both;
- `noise_to_signal` is strongly correlated with the roughness features;
- `peak_snr` is strongly negatively correlated with `noise_to_signal`.

Therefore, it is better to interpret the results by feature families instead of individual variables. A reduced and more interpretable feature set is:

- sampling geometry: `n_train`, `gap_90_train`, and `nearest_train_distance_mean`;
- morphology: `flat_fraction` and `width_50_pos`;
- roughness: `mean_abs_slope` and `median_crossing_rate`;
- noise: `noise_to_signal`.

2.5 Correlation Between Evaluation Metrics and Features

To investigate the relationship between light-curve characteristics and prediction performance, each feature was divided into approximately equal-sized groups using quantile binning (pd.qcut), corresponding to the low, medium, and high quantiles.

The results show that different GP metrics are sensitive to different feature families.

2.5.1 Signed Calibration Error

For `z_error`, the strongest correlations are with morphology and peak-related features:

- `flat_fraction`: -0.3447
- `width_50_pos`: 0.2988
- `fraction_train_near_peak`: -0.2337
- `peak_snr`: -0.2056

The negative correlation between `flat_fraction` and `z_error` means that objects with larger flat regions tend to have more negative `z_error`, i.e. they are more likely to be underconfident. This supports the visual observation that supernova-like curves, which often have sharp peaks and flat regions away from the peak, tend to produce underconfident single-band GP predictions.

2.5.2 CRPS

For normalized CRPS, the strongest correlations are with roughness, variability, and noise features:

- `mean_abs_slope`: 0.8203
- `total_variation`: 0.7480
- `flux_range_robust`: 0.6950
- `noise_to_signal`: 0.6480
- `median_crossing_rate`:0.4731

For each quantile group, the binned trend plots summarize the CRPS distribution using the median as the central value and the interquartile range (25th–75th percentiles) as the uncertainty band. They show a clear increase in median CRPS as `mean_abs_slope`, `noise_to_signal`, and `median_crossing_rate` increase. This indicates that rapidly varying, noisy, or oscillatory curves are harder for the single-band GP to predict.

However, the highest-complexity bins have large uncertainty. In these bins, the mean CRPS is often much larger than the median CRPS, indicating that a few extreme object-band cases dominate the mean. Therefore, median and interquartile range are more reliable summaries than mean and standard deviation for these skewed cases.

2.5.3 KS-PIT

For KS-PIT, the strongest correlations are with sampling geometry:

- `n_train`: -0.3900
- `gap_90_train`: 0.3558
- `nearest_train_distance_mean`: 0.3218
- `median_train_cadence`: 0.3156

Since lower KS-PIT is better, this means that more training points improve PIT calibration, while larger gaps and larger train-test distances worsen it. This suggests that PIT-based calibration is mainly sensitive to how well the test points are supported by nearby training observations.

2.6 Summary of Findings

The current single-band GP feature analysis suggests the following interpretation:

This is useful because it separates different failure modes. Roughness features strongly explain normalized CRPS, but they do not necessarily determine whether the GP is overconfident or underconfident. The direction of miscalibration is more related to transient morphology, while PIT-based calibration is more related to sampling geometry.

Metric	Main associated feature family	Interpretation
`z_error`	morphology and peak shape	direction of miscalibration
CRPS	roughness, noise, and variability	prediction difficulty
KS-PIT	sampling geometry	global calibration uniformity

3 Benefits of the Feature-Based Analysis

The feature-based analysis has several benefits.

First, it provides failure-mode diagnosis. Instead of only saying that one class performs worse than another, we can say which properties of the light curve are associated with poor GP performance. For example, rough and noisy curves have larger CRPS, while sparse sampling and large train-test distances worsen KS-PIT.

Second, it helps identify candidates for feature-conditioned calibration. A global uncertainty correction may be insufficient because different regimes show different calibration behavior. For example, transient-like curves with large flat regions may need different treatment from noisy, rapidly varying curves.

Third, it provides a basis for comparison with multi-band GP. If multi-band GP improves performance, we can ask whether the improvement is stronger in particular feature regimes, such as sparse sampling, poor peak coverage, or high noise-to-signal ratio.

4 Relation to Post-Hoc Calibration

The current feature analysis is diagnostic. Specifically, a strong correlation between `mean_abs_slope` and CRPS tells us that rough curves are harder to predict, but it does not directly tell us how to rescale the GP uncertainty.

For post-hoc calibration, we need signed calibration information such as `z_error` or empirical coverage error. A simple feature-conditioned calibration method would estimate a variance scaling factor within each feature group:

$$\sigma_{\mathrm{cal}} = s_g \sigma_{\mathrm{GP}},$$

where s_g is estimated from the standardized residuals in group g. If $s_g > 1$, the original GP is overconfident in that group and the predictive standard deviation should be enlarged. If $s_g < 1$, the GP is underconfident and the predictive standard deviation should be reduced.

However, for a valid calibration procedure, the features used for calibration should be available before seeing the test labels. Therefore, train-only versions of the features should be used for actual post-hoc calibration. Features computed

from the complete observed curve are useful for diagnosis, but they may leak information from the held-out test data if used directly for calibration.

5 Next Steps

The next steps are:

- Inspect the extreme object-band cases in the highest roughness and flat-fraction bins to determine whether they are valid difficult cases or preprocessing artifacts.
- Repeat the feature analysis using class-stratified plots, especially within the SNIa class, to check whether the observed trends remain within one class.
- Add GP-internal features such as optimized length scale and predictive standard deviation.
- Repeat the same feature-based analysis for multi-band GP and compare whether multi-band information reduces the feature dependence observed in single-band GP.
- For MOGP, it's important to perform band ablation to study band and cross-band information transfer.

In particular, the comparison with multi-band GP should investigate whether the effect of sampling sparsity, peak coverage, and roughness is weaker when information from other bands is available. This would help determine whether multi-band GP improves only average performance, or whether it specifically helps in difficult feature regimes.